



CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN ADVANCED PROFICIENCY EXAMINATION®

PHYSICS

UNIT 2 – Paper 02

2 hours 30 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of SIX questions in TWO sections. Answer ALL questions.
2. For Section A, write your answers in the spaces provided in this booklet.
3. For Section B, write your answers in the spaces provided at the end of each question in this booklet.
4. All working **MUST** be shown.
5. You may use a silent, non-programmable calculator to answer questions, but you should note that the use of an inappropriate number of figures in answers will be penalized.

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C.E.S.E.C.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

NOTHING HAS BEEN OMITTED.

LIST OF PHYSICAL CONSTANTS

Speed of light in free space	c	=	$3.00 \times 10^8 \text{ m s}^{-1}$
Permeability of free space	μ_0	=	$4\pi \times 10^{-7} \text{ H m}^{-1}$
Permittivity of free space	ϵ_0	=	$8.85 \times 10^{-12} \text{ F m}^{-1}$
			$\frac{1}{4\pi \epsilon_0} = 9.0 \times 10^9 \text{ m F}^{-1}$
Elementary charge	e	=	$1.60 \times 10^{-19} \text{ C}$
Planck's constant	h	=	$6.63 \times 10^{-34} \text{ J s}$
Unified atomic mass constant (Energy equivalence:	u $1u$	= =	$1.66 \times 10^{-27} \text{ kg}$ 931 MeV/c^2)
Rest mass of electron	m_e	=	$9.11 \times 10^{-31} \text{ kg}$
Rest mass of proton	m_p	=	$1.67 \times 10^{-27} \text{ kg}$
Acceleration due to gravity	g	=	9.81 m s^{-2}
1 Atmosphere	Atm	=	$1.00 \times 10^5 \text{ N m}^{-2}$
Avogadro's constant	N_A	=	$6.02 \times 10^{23} \text{ per mole}$

GO ON TO THE NEXT PAGE

SECTION A

Answer ALL questions.

Write your answers in the spaces provided in this booklet.

1. (a) With the aid of a diagram, derive the formula for the equivalent resistance of two resistors in parallel.

[6 marks]

- (b) Figure 1A is a potential divider circuit which splits up the 32 V from the battery into voltages, V_1 and V_2 . Figure 1B is the equivalent circuit to Figure 1A.

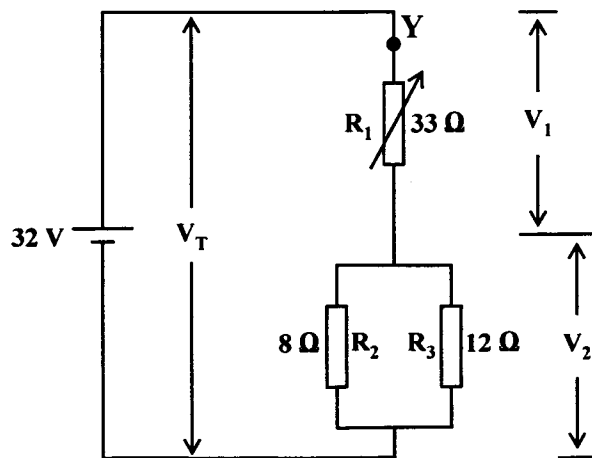


Figure 1A

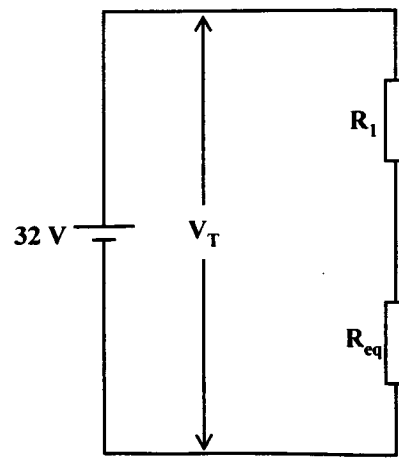


Figure 1B

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- (i) Calculate the equivalent resistance, R_{eq} in Figure 1B.

[2 marks]

- (ii) Determine the voltage, V_2 .

[1 mark]

- (c) The calculations above assume that the output resistance of the 32 V battery is zero. In practice this may not be so. In order to determine the internal resistance of the battery, an ammeter is inserted in the circuit at Y and the resistance of R_1 is reduced.

The terminal voltage, V_T , of the battery is monitored using a high resistance digital voltmeter. The readings of the ammeter (I) and the voltmeter (V_T) are recorded and tabulated. These are shown in Table 1 below.

TABLE 1

V_T/V	I/A
30	0.8
28	1.6
26	2.5
24	3.3
22	4.1

- (i) Use the results above to plot a graph of I against V_T on the grid provided on page 7. **[3 marks]**
- (ii) The slope, S , of the graph is related to the internal resistance, r , by $S = -\frac{1}{r} \Omega^{-1}$. Find S and hence determine the internal resistance, r , of the battery.

[3 marks]

Total 15 marks

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2. Figure 2 shows the gain-frequency curve (the gain-frequency response) of a general purpose operational amplifier (op-amp).

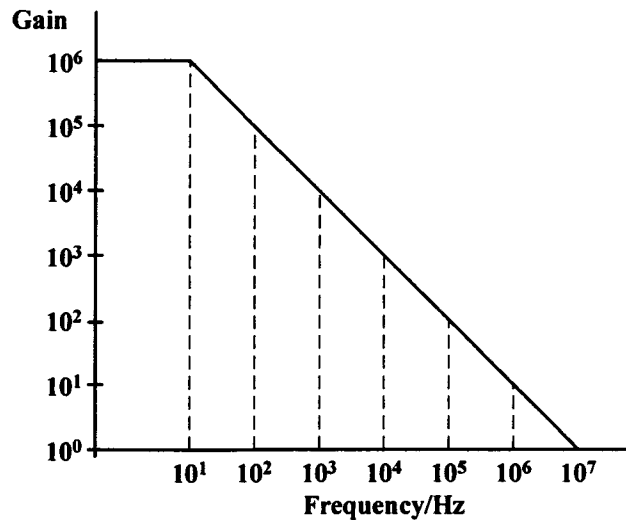


Figure 2

- (a) Use the graph to determine the
- (i) open loop d.c. gain of the amplifier
 - (ii) unity gain bandwidth of the amplifier
 - (iii) open loop bandwidth of the amplifier.

[1 mark]

[1 mark]

[1 mark]

GO ON TO THE NEXT PAGE

- (b) The op-amp with frequency response shown in Figure 2 is to be used as a non-inverting amplifier with the feedback resistance $R_f = 330 \text{ k}\Omega$ and input resistance $R_i = 3.3 \text{ k}\Omega$.

- (i) Draw the circuit diagram for this amplifier and label the resistors with their values.

[3 marks]

- (ii) Calculate the gain of the amplifier.

[2 marks]

- (iii) Use the open loop gain-frequency curve in Figure 2 to determine the closed loop bandwidth of this amplifier.

[2 marks]

- (iv) Deduce the gain of the amplifier when the input resistance is made infinitely large (so that it can be effectively removed from the circuit).

[2 marks]

- (v) What would be the input impedance of the resulting circuit?

[1 mark]

- (vi) Identify the practical application for the circuit and state why it is suited for this application.

[2 marks]

Total 15 marks

GO ON TO THE NEXT PAGE

3. (a) Describe the phenomenon of 'photoelectric emission'.

[3 marks]

- (b) Identify the property of a light source that would make it suitable for a demonstration of this phenomenon and give a reason why this property makes it suitable.

[2 marks]

- (c) Calculate the energy (in eV) of a light photon of wavelength 200 nm.

[3 marks]

- (d) The light source in Part (c) is used in a photoemission experiment with a metal whose work function is, $\Phi = 4.6 \text{ eV}$. Calculate the maximum velocity of the electrons ejected from the surface.

[3 marks]

- (e) The data in Table 2 were obtained for a photoemission experiment using Caesium.

TABLE 2

Stopping Potential V_s/V	Wavelength λ/nm
2.8	253
2.3	280
1.9	313
1.3	365
1.0	405

- (i) On the grid provided on page 13, plot a graph of V_s against λ . **Ensure that the λ scale extends to 550 nm.** [3 marks]
- (ii) From the graph determine the cut-off wavelength for Caesium.

[1 mark]

Total 15 marks

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1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200	201	202	203	204	205	206	207	208	209	210	211	212	213	214	215	216	217	218	219	220	221	222	223	224	225	226	227	228	229	230	231	232	233	234	235	236	237	238	239	240	241	242	243	244	245	246	247	248	249	250	251	252	253	254	255	256	257	258	259	260	261	262	263	264	265	266	267	268	269	270	271	272	273	274	275	276	277	278	279	280	281	282	283	284	285	286	287	288	289	290	291	292	293	294	295	296	297	298	299	300	301	302	303	304	305	306	307	308	309	310	311	312	313	314	315	316	317	318	319	320	321	322	323	324	325	326	327	328	329	330	331	332	333	334	335	336	337	338	339	340	341	342	343	344	345	346	347	348	349	350	351	352	353	354	355	356	357	358	359	360	361	362	363	364	365	366	367	368	369	370	371	372	373	374	375	376	377	378	379	380	381	382	383	384	385	386	387	388	389	390	391	392	393	394	395	396	397	398	399	400	401	402	403	404	405	406	407	408	409	410	411	412	413	414	415	416	417	418	419	420	421	422	423	424	425	426	427	428	429	430	431	432	433	434	435	436	437	438	439	440	441	442	443	444	445	446	447	448	449	450	451	452	453	454	455	456	457	458	459	460	461	462	463	464	465	466	467	468	469	470	471	472	473	474	475	476	477	478	479	480	481	482	483	484	485	486	487	488	489	490	491	492	493	494	495	496	497	498	499	500	501	502	503	504	505	506	507	508	509	510	511	512	513	514	515	516	517	518	519	520	521	522	523	524	525	526	527	528	529	530	531	532	533	534	535	536	537	538	539	540	541	542	543	544	545	546	547	548	549	550	551	552	553	554	555	556	557	558	559	560	561	562	563	564	565	566	567	568	569	570	571	572	573	574	575	576	577	578	579	580	581	582	583	584	585	586	587	588	589	590	591	592	593	594	595	596	597	598	599	600	601	602	603	604	605	606	607	608	609	610	611	612	613	614	615	616	617	618	619	620	621	622	623	624	625	626	627	628	629	630	631	632	633	634	635	636	637	638	639	640	641	642	643	644	645	646	647	648	649	650	651	652	653	654	655	656	657	658	659	660	661	662	663	664	665	666	667	668	669	670	671	672	673	674	675	676	677	678	679	680	681	682	683	684	685	686	687	688	689	690	691	692	693	694	695	696	697	698	699	700	701	702	703	704	705	706	707	708	709	710	711	712	713	714	715	716	717	718	719	720	721	722	723	724	725	726	727	728	729	730	731	732	733	734	735	736	737	738	739	740	741	742	743	744	745	746	747	748	749	750	751	752	753	754	755	756	757	758	759	760	761	762	763	764	765	766	767	768	769	770	771	772	773	774	775	776	777	778	779	780	781	782	783	784	785	786	787	788	789	790	791	792	793	794	795	796	797	798	799	800	801	802	803	804	805	806	807	808	809	810	811	812	813	814	815	816	817	818	819	820	821	822	823	824	825	826	827	828	829	830	831	832	833	834	835	836	837	838	839	840	841	842	843	844	845	846	847	848	849	850	851	852	853	854	855	856	857	858	859	860	861	862	863	864	865	866	867	868	869	870	871	872	873	874	875	876	877	878	879	880	881	882	883	884	885	886	887	888	889	890	891	892	893	894	895	896	897	898	899	900	901	902	903	904	905	906	907	908	909	910	911	912	913	914	915	916	917	918	919	920	921	922	923	924	925	926	927	928	929	930	931	932	933	934	935	936	937	938	939	940	941	942	943	944	945	946	947	948	949	950	951	952	953	954	955	956	957	958	959	960	961	962	963	964	965	966	967	968	969	970	971	972	973	974	975	976	977	978	979	980	981	982	983	984	985	986	987	988	989	990	991	992	993	994	995	996	997	998	999	1000
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SECTION B

Answer ALL questions.

Write your answers in the spaces provided at the end of each question.

4. (a) Figure 3 shows a model of an electron gun similar to that found in older television sets and oscilloscopes. Electrons from a thermionic emitter are accelerated to high velocities by the potential difference between the plates and pass through a small hole in the negative plate.

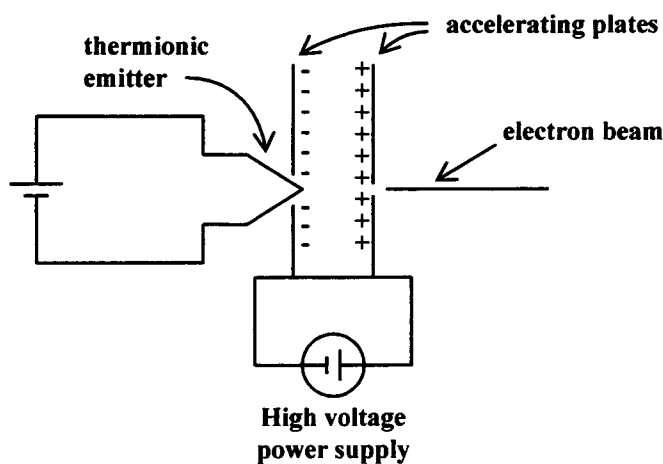


Figure 3

- (i) By equating the gain in kinetic energy of the electrons to the work done in accelerating them, show that the velocity, v_e , of the emerging beam electrons is given by:

$$v_e = \sqrt{\frac{2e V_p}{m_e}}$$

where V_p is the potential difference between the plates.

- (ii) Calculate the velocity of the electrons in the emerging electron beam when $V_p = 25 \text{ kV}$. [4 marks]

GO ON TO THE NEXT PAGE

- (b) Figure 4 shows a long solenoid which has 1000 turns per metre and carries a current of 10 A.

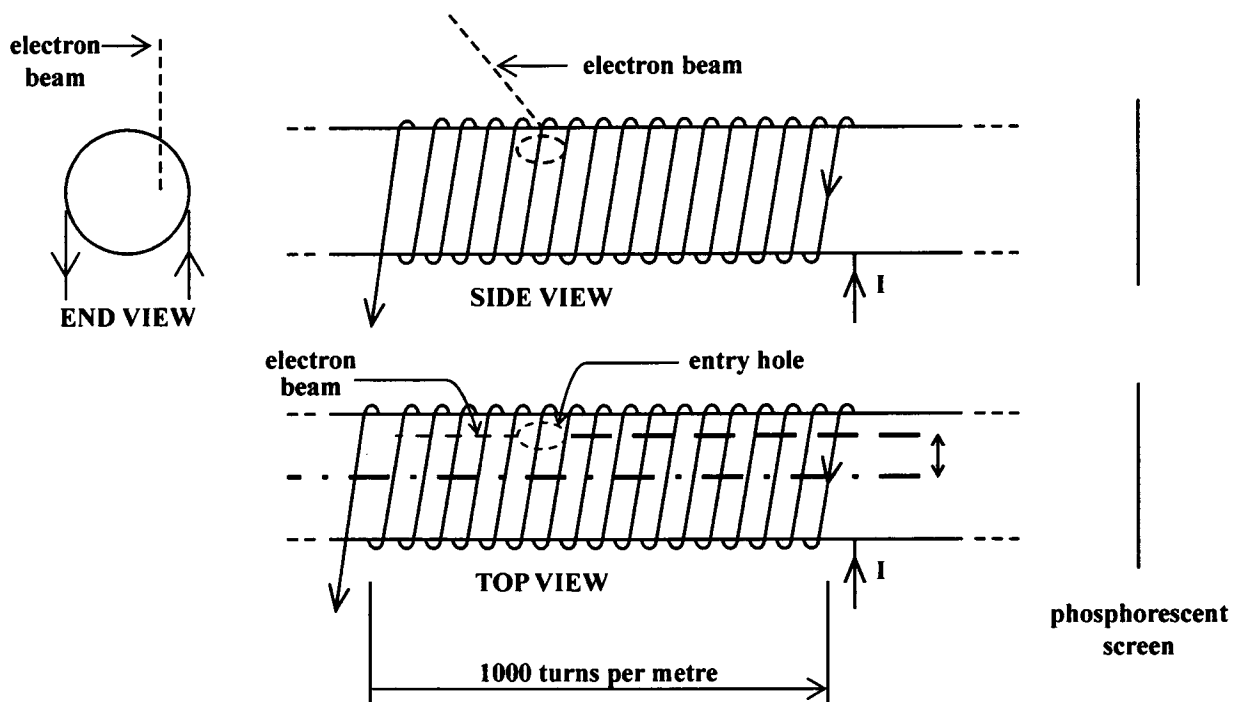


Figure 4

- (i) Calculate the magnitude and direction of the magnetic field inside the solenoid.
[3 marks]
- (ii) The solenoid is placed with its long axis horizontal so that a tiny hole (shown exaggerated in the diagram) faces upward and is displaced from the axis by a radial distance X . An electron beam is directed through the hole as shown in Figure 4.
 - a) Explain the effect on the electron's **horizontal** velocity component, resulting from its interaction with the B-field.
 - b) Describe the effect on the electron's **vertical** velocity component.
 - c) Describe the resultant motion of the electrons and the nature of the image produced on the phosphorescent screen when the B-field is large enough to prevent the electrons from touching the interior of the solenoid.

[8 marks]

Total 15 marks

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[illegible]

Write the answer to Question 4 here.

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5. Some modern cars incorporate an electronic system for turning on the headlights automatically whenever the light levels of the surroundings fall below a certain threshold. Other specifications of this type of system are:

- the lights can only come ON if the ignition switch S1 (logic A) is ON
- the headlights AUTO switch S2 (logic B) must be ON for the lights to come ON
- lighting level (logic C) below threshold is to be taken as logic 0

- (a) Use logic Y to represent the output of this system and draw up the truth table for the operation of the headlight automatic switching circuit. Write the logic equation represented by this truth table. **[3 marks]**

- (b) The circuit shown in Figure 5 is proposed for implementing the headlight auto switching system.

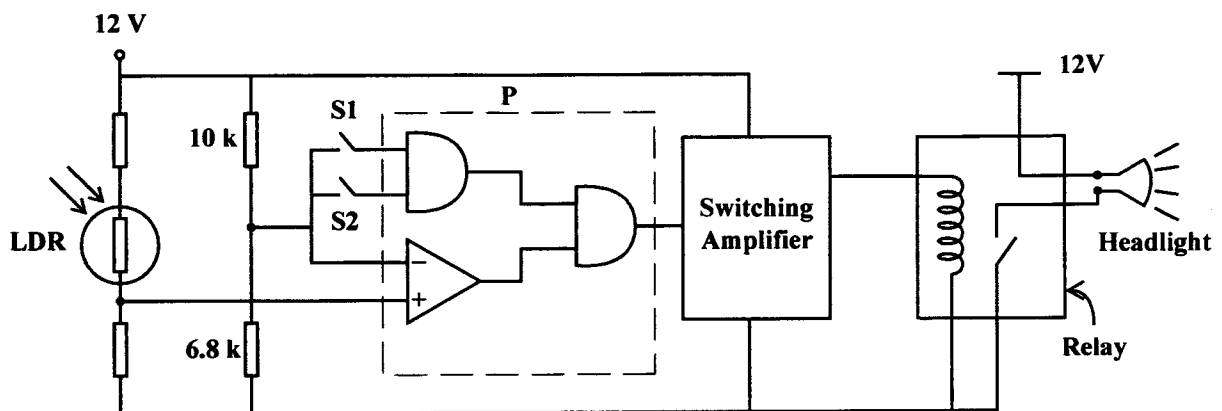


Figure 5

- (i) In this circuit, the open loop operational amplifier is used to compare the surrounding lighting level with a preset value. Explain how this is done.
- (ii) Justify the need for the light dependent resistor (LDR) in the circuit.
- (iii) Describe how a relay functions, stating why this device is necessary in the circuit. **[7 marks]**
- (c) (i) Replace the circuit in the dashed box, P, with one consisting of NAND gates only.
- (ii) The 7400 chip has 4 NAND gates. How many 7400 chips are required for the NAND-equivalent circuit drawn in Part c (i) ? **[5 marks]**

Total 15 marks

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[illegible][illegible]

6. (a) A radioactive sample initially contains N_0 radioactive particles. Its decay constant is λ and it decays to N particles in time, t . Write the equation describing this decay. [1 mark]

- (b) A capacitor, C , initially charged to voltage, V_0 , will discharge through a resistance, R , according to the equation $V = V_0 \exp(-t/\tau)$, where τ is called the time constant.

- (i) Write the equation expressing τ , in terms of R and C .
- (ii) If the capacitive discharge is used to model radioactive decay, where voltage is the analogue of number of particles, what would be the relationship between time constant, τ and decay constant, λ ?
- (iii) Given that $t_{1/2}$ is the half-life of a radioactive sample, use the decay equation from

Part (a) to show that $t_{1/2} = \frac{0.69}{\lambda}$. [5 marks]

- (c) The circuit shown below is used to model the decay of the radioactive isotope Francium-221, which has a half-life of 288 s. S_1 is initially switched to terminal a .

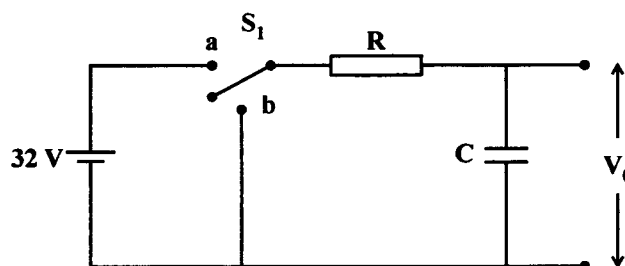


Figure 6

- (i) Describe the change in the voltages across R and C , both of which initially had no voltage across them.
- (ii) After a sufficiently long time, the voltages across R and C become fairly stable and S_1 is then switched to terminal b , so that the voltage across C models the decay of the Francium-221 isotope. Using a value of $R = 1 \text{ M}\Omega$, calculate the value of C needed to model the radioactive decay of Francium-221. [6 marks]

- (d) (i) State what is meant by the term 'activity' of a radioactive nucleus and write its expression in mathematical form.

- (ii) A sample of Francium-221 contains 10^{12} atoms. Calculate its activity. [3 marks]

Total 15 marks

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Write the answer to Question 6 here.

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IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.